SHAIKH ABDUL ZUBAIR ABDUL KABIR

Full-Stack & Mobile Developer | Edge AI & Offline Mesh Systems Specialist

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🎓 BCA Final Year — SRK Institute (2026/2027)

PROFESSIONAL SUMMARY

High-velocity and versatile **Software Developer & BCA Final Year Student** at **SRK Institute of Business Administration and Computer Education**. Creator and architect of **55+ end-to-end applications** spanning Native Android, multi-app Flutter ecosystems, Full-Stack web platforms, Local Edge AI, and Game Engines (Unity 3D / Godot 4). Completed a dedicated **Software Engineering Internship (50-day development track with 83.7%+ productive rating)**. Proven track record in offline-first architectures, peer-to-peer mesh communications (BLE/Wi-Fi Direct), cryptographic verification (Ed25519), and browser automation agents.

CORE TECHNICAL SKILLS

Languages	Java, Kotlin, Dart, Python, TypeScript, JavaScript, SQL, C#, C++, GDScript, C, HTML5/CSS3
Mobile Development	Android Native (Java/Kotlin, Jetpack Compose, Coroutines, AccessibilityService, NanoHTTPD), Flutter (Multi-App Suites, SQLite, Dio), React Native
Web & Backend	Node.js, Express.js, React 18, Vite, Tailwind CSS, REST APIs, WebSockets, Socket.IO, Leaflet.js, Monaco Editor, xterm.js
AI & Edge Computing	Local LLMs (Llama, Ollama), Android On-Device Inference, PyTorch Custom Neural Networks, Agentic Loops (RAG, Autonomous Planners), Playwright Automation
Databases & Cloud	Supabase (PostgreSQL), MongoDB, SQLite, Firebase (Firestore, Cloud Messaging FCM, Auth), Render Cloud
Networking & Security	BLE (Bluetooth Low Energy), Wi-Fi Direct, P2P Mesh Routing, Ed25519 Cryptographic Signatures, Dynamic Nonce QR, PathGuard Sandbox
Tools & Game Engines	Android Studio, VS Code, Visual Studio, Git/GitHub, Postman, Unity 3D (NavMesh, URP), Godot Engine 4, Arduino IDE, FFmpeg

WORK EXPERIENCE & INTERNSHIP

Software Developer Intern — Full-Stack & Engineering — **50 Calendar Days (83.72% Productive Efficiency Rate)**

- Contributed to end-to-end software module development, API integration, and architectural debugging across distributed codebases.
- Engineered offline caching, database schema improvements, and frontend performance optimizations.
- Collaborated across agile remote sprints with structured daily tracking, issue resolution, and system reporting.

COMPLETE PROJECT PORTFOLIO (ALL 55+ PROJECTS)

1. CAMPUS, ENTERPRISE & FULL-STACK SYSTEMS

SRK Campus Connect — Campus Presence & Faculty Coordination System

Flutter, Node.js, TypeScript, Express, MongoDB, Socket.IO, Ed25519

- Architected an **offline-first campus ecosystem** unifying 3 Flutter client applications (Student App, Faculty Dashboard, Admin Console) and a unified Node.js/TypeScript backend for SRK Institute.
- Engineered **dynamic 15-second rolling QR codes** with rotating nonces and Ed25519 cryptographic signatures to eliminate proxy attendance and replay attacks.
- Implemented an autonomous **P2P peer-forwarding gateway election** protocol allowing offline students to securely relay attendance packets through connected peers.
- Built a multi-factor **Presence Confidence Engine** calculating weighted evidence scores (0–100%) using BLE proximity and cryptographic timestamps.

ProjectFlow Intelligence — Enterprise Operations Platform

Node.js, Express, React 18, Tailwind CSS, Supabase PostgreSQL, Firebase FCM, Android

- Enterprise task intelligence platform deployed live on Render Cloud with Supabase PostgreSQL cloud persistence.
- Integrated **Firebase Cloud Messaging (FCM)** for instantaneous push notification routing across web management consoles and native Android devices.
- Engineered an offline-first synchronization engine ensuring zero data loss during unstable field network conditions.

MasiPay (MasiCanteen)

TypeScript, React Native (Android), Node.js, React Web

College canteen digital ordering and live token lifecycle management platform for campus crowds.

Smart Expenses Tracker (divya)

Android (Java/Kotlin), React.js Admin, Node.js, SQLite

Offline-first personal finance tracker with expense categorization, monthly threshold alerts, and cloud sync.

Follow Or Not (follow or not app)

React 18, Vite, TypeScript, Tailwind CSS

Social audience analytics application tracking follower mutuals, unfollowers, and relationship metrics.

Full-Stack Social Network (js\socialmedia)

React.js, Node.js, Express, MongoDB, WebSockets

Complete social network platform with user profiles, photo feeds, live post interactions, and real-time chat.

Modular MVC Backend (js\backend)

Node.js, Express, MongoDB, JWT Middleware

Enterprise REST API architecture featuring modular controllers, authentication middleware, and validated schemas.

College Hub Portal (collage-hub)

React.js, Node.js, WebSockets

Campus collaboration hub for notes sharing, exam schedules, and inter-departmental notices.

2. OFFLINE-FIRST, P2P MESH & NATIVE ANDROID SYSTEMS

AapdaSetu / NearMeshChat — Disaster Response & Offline Mesh Network

Android Native (Java), BLE Radar, Local Llama Edge AI, React.js, Express, Leaflet

- Engineered an emergency disaster communication network operating completely independent of active cellular towers or internet access.
- Built an **Android client featuring a real-time BLE Radar scanner**, peer distance estimation, and mesh packet store-and-forward routing.
- Integrated **on-device local LLM edge inference (Llama)** directly on Android to provide medical triage, survival guidance, and emergency response instructions without network connectivity.
- Created a centralized **Emergency Operations Command Web Dashboard** with real-time incident mapping using Leaflet and bidirectional Socket.IO dispatch.

Nexus (MeshLinkOffline & PocketAI Server) — Edge AI HTTP Engine

Android Native (Kotlin/Java, 61 source files), Local AI Engine, NanoHTTPD

- Mobile edge-computing system hosting a local on-device HTTP microserver (`PocketAiHttpServer`) and an embedded AI inference engine.
- Allows nearby devices on the local Wi-Fi/Hotspot network to query AI knowledge bases and retrieve computed answers without active internet connectivity.

RecoverX (recover_apk)

Native Android (Kotlin), Low-Level Storage APIs

Deep storage explorer for forensics-informed file carving, deleted media recovery, and disk block inspection.

Couple Canvas (couple draw)

Android Native (Kotlin), Node.js WebSockets

Real-time dual-user collaborative drawing app with sub-50ms synchronized touch sketching and stroke vectorization.

Emergency Recorder (Fast Rec)

Kotlin, Jetpack Compose, AudioRecord API

Covert emergency audio recording utility with instant panic triggers, background operation, and encrypted storage.

Personal Life Vault (sthelp)

Android Native, AES-256 Encryption, Node.js Daemon

Hardware-backed encrypted vault for sensitive documents with stealth launch triggers and local PC synchronization.

P.RN Controller / Focus Guard

Android Native, AccessibilityService, DevicePolicyManager

Completely offline digital addiction blocker enforcing self-control policies with zero internet telemetry.

Z-Music Player

Android Native (Kotlin), ExoPlayer, MediaSession

High-fidelity music streaming and offline audio player with foreground background service and lock-screen controls.

myhelper & NearDrop Utilities

Android Native, Wi-Fi Direct, mDNS

Device cleanup analyzer identifying redundant cache files, and local zero-configuration wireless file drop utility.

Android Prototypes (MyApplication 1–4)

Java / Kotlin, Android SDK, CameraX, Sensors

Prototypes testing custom CameraX capture pipelines, sensor data fusion (gyroscope/accelerometer), and Room DB migrations.

3. AUTONOMOUS AI AGENTS & GENERATIVE MEDIA STUDIOS

Autonomous Web-Agent IDE — Local Natural-Language Software Generator

Python, React 18, TypeScript, Tailwind CSS, Monaco Editor, Playwright, xterm.js

- Built a local desktop IDE that transforms natural-language project requirements into full-stack, tested, and running code using web-based LLMs.
- Created headless browser orchestration with **Playwright to drive web LLMs** without requiring commercial paid API keys.
- Implemented **PathGuard security sandbox** preventing directory traversal and symlink attacks with strict SAFE/CAUTION/BLOCKED command policies.
- Integrated multi-tab **Monaco Editor**, live terminal streaming via **xterm.js**, task graph DAG, and automated Git checkpointing with loop-breaking error recovery.

AURA — Autonomous Local AI Research & Learning System

Python, Local LLMs, Vector Memory DB, Playwright, DuckDuckGo API

- Designed a self-directed cognitive engine implementing a continuous **Think → Plan → Execute → Verify → Reflect → Store** loop.
- Optimized for local consumer hardware (NVIDIA RTX 2050 4GB VRAM) for autonomous knowledge gap discovery using an internal "Curiosity Engine".
- Built a multi-tier memory hierarchy (Semantic, Episodic, Procedural, and Skill Memory) with automated experimentation in sandboxed sub-processes.

My Custom AI (my_ai)

Python, PyTorch, Custom Tokenizer, NumPy

Built and trained a custom neural language model from scratch (`train.py`, `chat.py`, custom tokenizer, embedding & attention layers).

AI Zubair LLM Research & Audit Suite

Python, Google Colab, Jupyter Notebook, PyTorch

Comprehensive LLM phase-1 training audits, hyperparameter benchmarks, and automated dataset inventory validation scripts.

My Helper Autonomous Agent

Python, RAG Pipeline, Browser Automation

Modular cognitive assistant with dedicated modules for brain routing, vector memory, RAG, and headless web research tools.

OmniDev AI System (ai)

Python, SQLite Vector DB, Godot Engine CLI

Autonomous software engineering agent with long-term memory capable of autonomously creating, testing, and debugging Godot games.

StickForge AI (stick man vid)

Python, Generative AI, PIL, OpenCV, FFmpeg

Automated animation pipeline converting scripts into stickman animations, synthesized speech, and video rendering.

AI Content Repurposing Studio

Python, React 18, TypeScript, Whisper AI, FFmpeg

Desktop studio converting long-form videos into high-retention 9:16 vertical shorts with automated caption timing and highlight detection.

Auto YT Generator (yt video maker)

Python, Automated Asset Pipeline, FFmpeg

Automated 2D production engine generating full 1080p animated YouTube videos from raw topic inputs with automated scene transitions.

4. GAME DEVELOPMENT & INTERACTIVE GRAPHICS (GODOT & UNITY ENGINES)

Draw Your Army: WarForge (game 11)

Godot Engine 4.6.2, GDScript

Tactical RTS game combining freehand soldier sketching mechanics with real-time battlefield pathfinding and combat AI.

Minecraft 3D Block & Voxel Engine

Godot Engine, GDScript, 3D Shaders

3D voxel sandbox featuring procedural block placement, terrain generation, destruction physics, and player collision.

Full-Stack RPG Game Suite (game rpg)

Godot Engine, GDScript

Complete action RPG featuring character stats, combat state machines, enemy AI, inventory slots, and dungeon transitions.

Isekai World Adventure (iseaki_world)

Godot Engine, GDScript

Story-driven fantasy adventure game with interactive NPC dialog trees, quest progression, and world exploration.

Shadow Kingdom (shadow-kingdom)
Godot Engine, GDScript, Pixel Art

2D pixel-art dungeon adventure game featuring finite-state machines, custom tilemap collision physics, and enemy wave spawners.

Unity 3D FPS Game (My project / 1)
Unity 3D, C#, NavMesh AI, URP

First Person Shooter with NavMesh AI pathfinding, raycast shooting mechanics, particle impacts, and HUD overlay.

Slikj Water Physics Simulation
Unity 3D, C#, WaterSystem Shaders

Interactive 3D water physics simulation with realistic wave displacement, buoyancy calculations, and reflection shaders.

Yorus 3D Interactive World
Unity 3D Engine, C#

3D environment exploration featuring atmospheric lighting, custom collision detection, and third-person camera controls.

Arcade Games Suite (Space Dodger, Flappy Bird, Snake)
Godot Engine, GDScript

Complete suite of retro games: 2D Arcade Space Dodger, Flappy Bird clone with custom physics, and classic grid-based Snake engine.

360 Journey (Interactive Panorama Viewer)
JavaScript (ES6+), HTML5 Canvas, CSS3

Lightweight 360-degree panoramic viewer with smooth touch-drag physics without external heavyweight frameworks.

5. EMBEDDED SYSTEMS, WEB PORTALS & UTILITIES

Arduino IoT Wire Communication System
Arduino C/C++, I2C / Two-Wire Interface

Embedded microcontroller programming utilizing the Wire library for I2C master-slave communication and sensor data acquisition.

College & School Web Portals
HTML5, CSS3, JavaScript

Institutional portals for courses, departments, faculty profiles, student enrollments, fee status, and attendance rosters.

Stopwatch & Calculator Suite
JavaScript, HTML5, CSS3

Millisecond-precision timer with lap splitting and full arithmetic calculation engine with keyboard event listeners.

Systems & Algorithms Repositories (repos)
C, C++, Node.js

Low-level algorithmic implementations: memory allocation, linked lists, sorting benchmarks, and console utilities in C and C++.

EDUCATION

Bachelor of Computer Applications (BCA) — Final Year
SRK Institute of Business Administration and Computer Education

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java/C++), Database Management Systems (RDBMS & NoSQL), Mobile Application Development, Web Technologies, Software Engineering, Computer Networks, Operating Systems, Operational Research.

Graduating 2026 / 2027

KEY ENGINEERING STRENGTHS

- **Prolific Builder (55+ Projects):** Exceptional throughput across native Android, full-stack, local AI, and game engines.
- **Zero-Trust Offline-First Architectures:** Specialized in zero-connectivity communications using BLE, Wi-Fi Direct, and cryptographic verification (Ed25519).

- **Edge AI Engineering:** Hands-on experience quantizing, hosting, and fine-tuning local LLMs directly on consumer laptops and Android smartphones.